Chuck Wang
(Preferred start date : September 8th, but I am flexible)

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EDUCATION

University of Illinois at Urbana-Champaign Expected Graduation: May 2028

- Bachelor of Science in Computer Science + Statistics (GPA: 4.0), Minor in Data Science
- Main Courses: Algorithms & Models of Computation, Data Structures, Computer Systems, Data Science Pipeline

TECHNICAL PROJECTS & INTERNSHIP EXPERIENCE

National Center for Supercomputing Applications (NCSA) *Champaign, Illinois*

Guinea Pig Simulator 2.0 - Simulation Software Engineering Project

SPIN Intern *April 2026 ~ Present*

- Designed a modular genetics simulation engine with configurable inheritance rules, mutation settings, genotype-to-phenotype mappings, and multi-generation breeding logic.
- Built a rule-driven architecture that separates simulation configuration from core engine logic, improving extensibility, maintainability, and testability.
- Developed data-processing pipelines to aggregate simulated offspring outcomes into population-level phenotype distributions, generation trends, and visualization-ready datasets.
- Implemented Python-based visualization workflows for analyzing inheritance patterns, mutation effects, and trait frequency changes across generations.

IDX Exchange *Boise, Idaho*

Full-Stack Development Project - Property Search Website

Software Engineer Intern *October 2025 ~ December 2025* https://chuck.calisearch.org/

- Built backend PHP API endpoints to support property search, geolocation logic, advanced filtering, and dynamic map based queries.
- Implemented Google Maps powered map views with interactive markers, location based filtering, and synchronized search results.
- Integrated MySQL backed data retrieval with server side filtering logic to support structured property search and map driven navigation.
- Developed and tested front end search and filtering features using JavaScript, modular components, and responsive UI patterns.
- Deployed a production property search application on a Linux based hosting environment using cPanel, PHP, MySQL, and Linux server tooling.

University of Illinois Champaign, Illinois

Flight Route Intelligence Platform —Database-Backed Full-Stack Web Application

Personal Project *May 2026 ~ Present*

- Designed a full-stack flight analytics platform for searching routes, comparing airline reliability, and analyzing delay/cancellation trends using real-world aviation datasets.
- Modeled 5+ normalized database entities including flights, airports, airlines, routes, and performance records with one-to-many and many-to-many relationships.
- Integrated Google Maps API to visualize airport locations and flight routes, connecting SQL-backed route data with an interactive geographic interface.
- Built CRUD workflows and SQL-driven backend features using joins, aggregations, stored procedures, and analytical queries to support route search and reliability insights.

Other Professional and Leadership Experiences

Siebel School of Computing and Data Science, College of Liberal Art & Science, University of Illinois
Champaign, Illinois

Course Assistant and LAS100 Intern *February 2025 ~ May 2025*

- Delivered weekly programming tutorials and supported 50+ students with debugging, assignment strategies, and core programming concepts.

SKILLS & INTERESTS

Programming: TypeScript, JavaScript, SQL, Java, Python, C++
Web & Backend: REST APIs, React, Git, Linux, PHP, Google Maps API
Data and Databases: Relational Databases, MySQL, PostgreSQL, Data Modeling, Data Validation, API Design
Machine Learning: Data Modeling, Pagination, Matplotlib, Seaborn, TensorFlow, scikit-learn, Statsmodels, Seaborn